Experiment - 4

Fitting the Probability Distributions

(Discrete and Continuous Distribution)

Aim:

The aim is to understand discrete and continuous probability distribution using R.

Description: -

Probability distributions describe how probabilities are distributed over the values of a random variable. They are broadly classified into discrete and continuous distributions. Fitting a probability distribution helps in modeling real-life data and making predictions.

Discrete Probability Distributions

A discrete probability distribution is defined for a random variable that can take only a finite or countable number of values.

Binomial Distribution

A random variable X is said to follow a Binomial distribution if it satisfies the following conditions:
1. The experiment consists of n identical trials.
2. Each trial results in one of two possible outcomes: success or failure.
3. The probability of success (p) remains constant.

Probability Mass Function: $P(X = x) = nC_x\, p^x q^{n-x}, x = 0,1,2, \cdots n,\ \ q = 1 - p.$

Procedure

- Input/Import the data set
- Determine the probabilities of the random variable using Binomial distribution in R
- Visualize the probability distribution using R functions

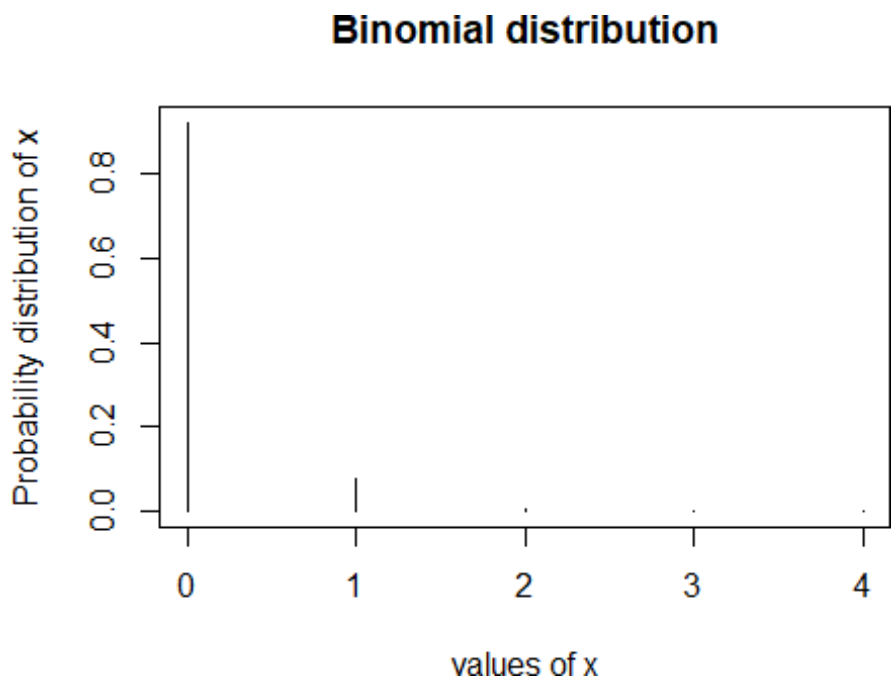
Problem-1:

Four coins are tossed simultaneously. What is the probability of getting (i) 2 heads (ii) at least 2 heads (iii) at most 2 heads (iv) Expectation of x (v) Variance of x (vi) Visualize the probability distribution (vii) 10th, 20th quantiles of binomial distribution (viii) How many heads will turn up with a probability of 0.25, when a single coin is tossed 51 times? (ix) Generate 8 random values from this distribution.

Code and Results:

```
# number of trials
n=4
n

## [1] 4

#probability of success
p=0.5
p

## [1] 0.5

# (i) probability of getting exactly 2 heads
dbinom(2,n,p)
## [1] 0.00230496

# (ii) probability of getting at least 2 heads
Sum(dbinom(2:4,n,p))
## [1] 0.00233648

#or
1-pbinom(1,n,p)
## [1] 0.00233648

# (iii) probability of getting at most 2 heads
Sum(dbinom(0:2,n,p))
## [1] 0.9999685

#or
pbinom(2,n,p)
## [1] 0.9999685

# (iv) Expectation of x
X=0:n
px=dbinom(x,n,p)
Ex=weighted.mean(x,px)
Ex
## [1] 0.08
```

```
# (v) Variance of x
Varx=weighted.mean(x*x,px)-(weighted.mean(x,px))^2
Varx
## [1] 0.0784

# (vi) Visualize probability distribution
plot(x, px, type=”h”, xlab=”values of x”, ylab=”Probability distribution
of x”, main= “Binomial distribution”)
```




```
# (vii) 10th, 20th quantiles of Bin(10, 1/2) distribution?
qbinom(0.1,10,1/2)
## [1] 1

qbinom(0.2,10,1/2)
## [1] 2

# (viii) How many heads will have a probability of 0.25 will come out when a
coin is tossed 51 times?
qbinom(0.25,51,1/2)
## [1] 23

# (ix) rbinom: This function generates required number of random values of given
probability from a given sample.

#Find 8 random values from a sample of 150 with probability of 0.4.
x<-rbinom(8,150,0.4)
print(x)
## [1] 50 59 63 51 52 66 52 62
```

Poisson Distribution

A discrete random variable X follows a Poisson distribution if it represents the number of events occurring in a fixed interval of time or space, and events occur independently.

Probability Mass Function:

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \quad x = 0,1,2,\cdots$$

Procedure

- Input/Import the data set
- Determine the probabilities of the random variable using Poisson distribution in R
- Visualize the probability distribution using R functions

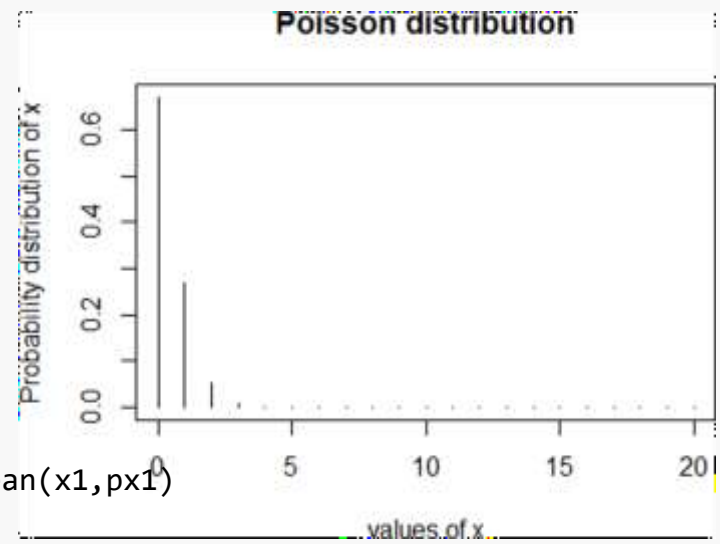
Problem-2:

A manufacturer of pins knows that 2% of his products are defective. If he sells pins in boxes of 20, find the number of boxes containing (i) at least 2 defective (ii) exactly 2 defectives (iii) at most 2 defective pins in a consignment of 1000 boxes (iv) plot the distribution (v) E(x) (vi) Variance of X? (vii) find the median of this distribution (viii) Generate 8 random values from this distribution.

Codes and Results:

```
#Poisson Distribution
# number of trails
m=20
m

## [1] 20

# probability of success
ps=0.02
# poisson parameter
lambda=m*ps
lambda

## [1] 0.4

#at least 2 defectives
p1=sum(dpois(2:m,lambda))
p1

## [1] 0.06155194

# (i) number of boxes containing at least 2 defectives
round(1000*p1)

## [1] 62

#exactly 2 defectives
p2=dpois(2,lambda)
p2

## [1] 0.0536256
```

```
# (ii) number of boxes containing exactly 2 defectives
round(1000*p2)

## [1] 54

#at most 2 defectives
p3=sum(dpois(0:2, lambda))
p3

## [1] 0.9920737

# (iii) number of boxes containing at most 2 defectives
round(1000*p3)

## [1] 992

# (iv) plot the distribution
x1=0:m
px1=dpois(x1,lambda)
plot(x1,px1,type="h",xlab="values of x",ylab="Probability distribution of
x",main="Poisson distribution")
```




```
#(v) E(x)
Ex1=weighted.mean(x1,px1)
Ex1

# (vi) variance of x
Varx1=weighted.mean(x1*x1,px1)-(weighted.mean(x1
,px1))^2 Varx1

# (vii) median of the given Poisson distribution

med=qpois(0.5,lambda)

# (viii) Generate 8 random numbers from the Poisson distribution

x=rpois(8,lambda)

print(x)
```

qpois( ) function is used to calculate the quantiles of a Poisson distribution. It takes two arguments: the probability at which to evaluate the quantile and the mean of the Poisson distribution.

For example, to calculate the 95th percentile of a Poisson distribution with a mean of 2.5, you can use the following code:

```
qpois(0.95, 2.5)
```

[1] 5

rpois( ) function is used to generate random numbers from a Poisson distribution. It takes one argument: the mean of the Poisson distribution.

rpois(q, lambda)

where, q: number of random numbers needed lambda: mean per interval rpois(2, 3)

##[1] 4 1

rpois(6, 6)

##[1] 9 2 9 9 1 10

Exercise (To be written in record notebook):

Problems-1 and 2

Continuous Probability Distributions

A continuous probability distribution is defined for a random variable that can take any value in a given interval.

Normal Distribution

A continuous random variable X is said to follow a Normal distribution with parameters μ and σ if its probability density function is:

$$f(x) = (1 / (\sigma\sqrt{(2\pi)})) \, e^{-}(x - \mu)^2 / (2\sigma^2)$$

The normal distribution is symmetric about the mean μ.

Procedure:

- Generating the data set
- Determine the probabilities of the random variable using Normal distribution in R
- Visualize the probability distribution using R functions

dnorm(x, mean = 0, sd = 1, log = FALSE)
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
qnorm(p, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
rnorm(n, mean = 0, sd = 1)

Arguments

x, q vector of quantiles.
p vector of probabilities.
n number of observations.
If length(n) > 1, the length is taken to be the number required. mean vector of means.
sd vector of standard deviations
log, log.p logical; if TRUE, probabilities p are given as log(p).
lower.tail logical; if TRUE (default), probabilities are P[X ≤ x] otherwise, P[X > x].

Problem:

A company finds that the time taken by one of its engineers to complete or repair job has a normal distribution with mean 20 minutes and S.D 5 minutes. State what proportion of jobs take:

i. Less than 15 minutes
ii. More than 25 minutes
iii. Between 15 and 25 minutes
iv. Plot the distribution
v. Table the distribution

Code and Results:

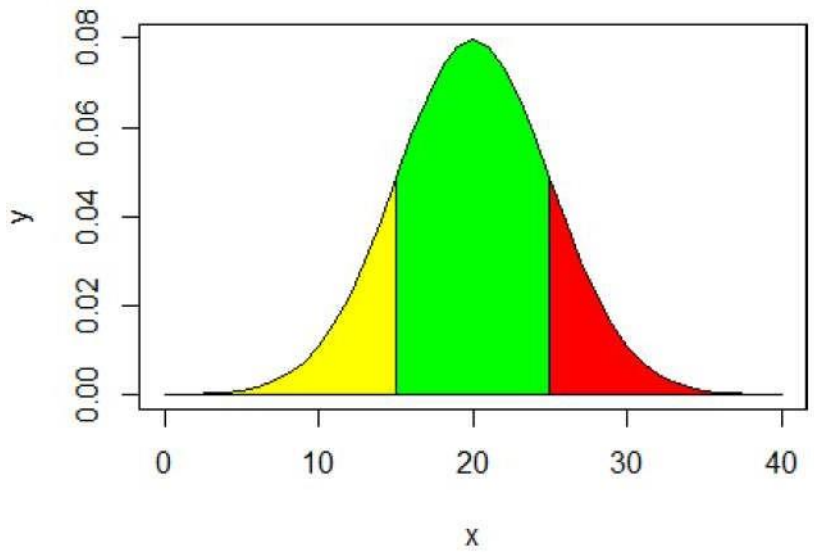
```
# Generating the data x
x=seq(0,40)
x

##  [1]  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22
23 24
## [26] 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

# find the density function of x
y=dnorm(x,mean=20,sd=5)
y

## [1] 2.676605e-05 5.838939e-05 1.223804e-04 2.464438e-04 4.768176e-04
## [6] 8.863697e-04 1.583090e-03 2.716594e-03 4.478906e-03 7.094919e-03
## [11] 1.079819e-02 1.579003e-02 2.218417e-02 2.994549e-02 3.883721e-02
## [16] 4.839414e-02 5.793831e-02 6.664492e-02 7.365403e-02 7.820854e-02
## [21] 7.978846e-02 7.820854e-02 7.365403e-02 6.664492e-02 5.793831e-02
## [26] 4.839414e-02 3.883721e-02 2.994549e-02 2.218417e-02 1.579003e-02
## [31] 1.079819e-02 7.094919e-03 4.478906e-03 2.716594e-03 1.583090e-03
```

```
## [36] 8.863697e-04 4.768176e-04 2.464438e-04 1.223804e-04 5.838939e-05
## [41] 2.676605e-05

# plot the normal distribution curve
plot(x,y,type='l')
# Proportion of jobs take less than 15 minutes
p1=pnorm(15,mean=20,sd=5)
p1
```

```
## [1] 0.1586553

x2=seq(0,15)
x2

##  [1]  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15

y2=dnorm(x2,mean=20,sd=5)
y2

## [1] 2.676605e-05 5.838939e-05 1.223804e-04 2.464438e-04 4.768176e-04
## [6] 8.863697e-04 1.583090e-03 2.716594e-03 4.478906e-03 7.094919e-03
## [11] 1.079819e-02 1.579003e-02 2.218417e-02 2.994549e-02 3.883721e-02
## [16] 4.839414e-02

polygon(c(0,x2,15),c(0,y2,0),col='yellow')
```

```
#Proportion of jobs take more than 25 minutes
p2=pnorm(40,mean=20,sd=5)-pnorm(25,mean=20,sd=5)
p2

## [1] 0.1586236

x1=seq(25,40)
x1

##  [1] 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

y1=dnorm(x1,mean=20,sd=5)
y1

## [1] 4.839414e-02 3.883721e-02 2.994549e-02 2.218417e-02 1.579003e-02
## [6] 1.079819e-02 7.094919e-03 4.478906e-03 2.716594e-03 1.583090e-03
## [11] 8.863697e-04 4.768176e-04 2.464438e-04 1.223804e-04 5.838939e-05
## [16] 2.676605e-05

polygon(c(25,x1,40),c(0,y1,0),col='red')

#Proportion of jobs take between 15 and 25 minutes
p3=pnorm(25,mean=20,sd=5)-pnorm(15,mean=20,sd=5)
p3

## [1] 0.6826895

x3=seq(15,25)
x3

##  [1] 15 16 17 18 19 20 21 22 23 24 25

y3=dnorm(x3,mean=20,sd=5)
```

```
y3
##  [1] 0.04839414 0.05793831 0.06664492 0.07365403  0.07820854 0.07978846
##  [7] 0.07820854 0.07365403 0.06664492 0.05793831  0.04839414

polygon(c(15,x3,25),c(0,y3,0),col='green')
```



```
# Probability distribution
data.frame(p1,p2,p3)

##          p1        p2        p3
## 1 0.1586553 0.1586236 0.6826895
```

The function `qnorm()`, which comes standard with R, aims to do the opposite: given an area, find the boundary value that determines this area.

For example, suppose you want to find that 85th percentile of a normal distribution whose mean is 70 and whose standard deviation is 3.

qnorm(0.85,mean=70,sd=3)

[1] 73.1093

The value 73.1093 is indeed the 85th percentile, in the sense that 85% of the values in a population that is normally distributed with mean 70 and standard deviation 3 will lie below 73.1093. In other words, if you were to pick a random member X from such a population, then

$$P(X < 73.1093)=0.85$$

It can be checked that this is correct by plugging 73.1093 into `pnorm()`:

```
pnorm(73.1093,,mean=70,sd=3)
The R function rnorm() is used for generating a vector of random numbers with a
normal distribution.
random_values <- rnorm(n = 25)
random_values
 ##[1] -0.53499657 -0.74971670 -0.22026809  1.54708162  0.39232568 -0.86422146
 ##[7]  0.36944164  0.49069040 -0.69812587 -0.13187081 -0.27770614  0.51801381
##[13]  0.57531833  0.11970355  0.84234079  1.28128005  0.92639749  1.20121957
##[19]  0.21812995 -0.02332619  1.06100364  0.55529338  0.45422197 -0.17755005
##[25] -1.44717709
```

Conclusion:

Suitable R functions have been explored to understand the discrete and continuous probability distributions.